COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 12 : SLO 1 — Tutorial: IEEE 754 Single Precision Encoding
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Section B: Fill in the Blanks (Understand)

- The IEEE 754 single precision format uses 32 bits in total.
- The exponent is stored with a bias of 127 in single precision.
- The sign bit is 0 for positive numbers and 1 for negative numbers.
- The actual exponent is calculated by subtracting the BIAS (127) from the stored exponent.
- The mantissa represents the FRACTIONAL part of the normalized binary number.

Section C: Activity — Encode -12.375 in IEEE 754 Single Precision

- Convert the magnitude: 12 = 1100, 0.375 = 0.011 → 1100.011
- Normalize: 1.100011 × 2^3
- Biased exponent: 3 + 127 = 130 = 10000010
- Sign bit: 1 (the number is negative)
- Sign: 1 Exponent (8-bit): 10000010 Mantissa (23-bit): 10001100000000000000000
- Final IEEE 754 Single Precision: 1 10000010 10001100000000000000000 (hex: C1460000)

Section D: Role-Based Play (Analyze)

- Roles: Decimal Converter (decimal → binary), Normalizer (1.x × 2^E form), Exponent Encoder (adds bias 127, converts to 8-bit), Sign Bit Decider (0 for +, 1 for -), Mantissa Filler (fraction bits into 23-bit field), Encoder Assembler (combines into the 32-bit word).
- Each role passes its result to the next stage, demonstrating how an FPU encodes a decimal value into IEEE 754 format step by step.